

Data Science Fundamentals

DS201 · Course Syllabus · Fall 2026

Instructor:	Prof. Amina Farouk
Email:	a.farouk@springfield-university.edu
Office Hours:	Mondays 1:00-3:00 PM, Room C310
Credit Hours:	4
Prerequisites:	CS101 or equivalent programming experience
Meeting Times:	Tue/Thu 1:00-2:50 PM, Lab Room C112

Course Description

This course provides a hands-on introduction to the data science workflow: acquiring, cleaning, analyzing, visualizing, and modeling data using Python's scientific computing stack (NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn). Students complete a capstone analysis on a real-world dataset of their choosing.

Learning Objectives

- Manipulate tabular data efficiently using Pandas
- Compute and interpret descriptive statistics
- Create effective visualizations to communicate findings
- Build, train, and evaluate basic supervised machine learning models
- Diagnose common issues such as overfitting and data leakage
- Present a data-driven story clearly to a non-technical audience

Required Materials

Students need a laptop capable of running Jupyter Notebook or JupyterLab, and should install Anaconda (Python 3.11 distribution) before Week 1. All datasets used in labs are provided on the course portal.

Grading Breakdown

Component	Weight	Notes
Weekly Lab Notebooks (10 total)	25%	Submitted via course portal
Midterm Exam	20%	Covers NumPy, Pandas, and statistics
Group Mini-Project	15%	Teams of 3, presented in Week 10
Capstone Project	30%	Individual, full data science pipeline

Participation	10%	Lab attendance and peer feedback
---------------	-----	----------------------------------

Grading Scale

Letter	Range
A	93-100
A-	90-92
B+	87-89
B	83-86
B-	80-82
C	70-79
F	Below 70

Course Policies

Lab notebooks are due at the start of the next lab session; late submissions lose 15% per day up to 3 days, after which they are not accepted. The capstone project must use a dataset with at least 500 rows and must not have been used in another course.

Collaboration is encouraged on labs, but the capstone project and exams must reflect individual work. Use of AI coding assistants must be disclosed in a short methods note for the capstone.

Weekly Topics Overview

Week	Topic
1	Data science workflow, Jupyter, NumPy basics
2	NumPy arrays, broadcasting, vectorization
3	Pandas Series and DataFrames
4	Data cleaning: missing values, duplicates, types
5	Descriptive statistics and distributions
6	Data visualization with Matplotlib and Seaborn
7	Midterm Exam
8	Intro to machine learning concepts
9	Linear and logistic regression

10	Group mini-project presentations
11	Decision trees and k-NN
12	Model evaluation and cross-validation
13	Capstone lab work
14	Capstone presentations